

Architecture > Population: A Research Program for the Inter-Agent Mesoscale

Hakvin Vosteen

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Abstract. The thirteen working pieces in this corpus repeat a single methodological move: a phenomenon ordinarily read as moral, cultural, or characterological is rewritten as the equilibrium of a mechanism with a threshold. We state the move as a research program organised around three claims. First, what looks like a property of *people* is usually a property of a *mechanism*; the recurrent objects are operators, sufficient statistics, and threshold games. Second, architecture beats population: replace the users while holding the mechanism fixed and the equilibrium is preserved; replace the mechanism and the equilibrium changes. Third, the scale at which most of contemporary subjective experience is decided — the inter-agent mesoscale — is methodologically nearly empty compared to the scales below the human (atoms, cells, neurons) and above (climate, markets-as-aggregate). The marginal return on formalising one previously unmodelled inter-agent mechanism is therefore large. We address the standard reflexive objection (modelling perturbs the modelled system) by distinguishing three modes of impossibility — information-theoretic, reflexive, and conceptual — and showing, with anchors in the corpus, that each admits a different attack.

1. The move. Each piece in this corpus takes a folk-psychological reading of a phenomenon and replaces it with a mechanism. *Burnout Is Not Too Much Work* replaces a moral reading (weakness, lack of grit) with the bifurcation of a friction-output ratio at $\rho^* = 1$. *Ghosting Isn't Cowardice* replaces a generational or character reading with the subgame-perfect equilibrium of a sequential exit game indexed by a silence-cost ratio σ_R . *ADHD Is Not a Deficit* replaces a deficit reading with a different attractor in a phase space whose stationary distribution is bimodal under a free-energy generative model. *Personality Is Not Who You Are* replaces a vector of traits with a four-dimensional cost tensor under a context operator. *Cartel or Creative Destruction?* replaces concentration-index conflation with eigenvector asymmetry. The move is the same in each case: take what reads as a property of the agent, and locate it in the structure the agent inhabits.

Two claims sit underneath the move.

Claim 1 (Mechanism, not person). *Phenomena that read as moral, cultural, or characterological are typically equilibria of mechanisms with thresholds. The dominant variable is not the agent's interior; it is the cost structure, signal architecture, or selection rule under which the agent acts.*

Claim 2 (Architecture, not population). *Hold the mechanism fixed and replace the population: the equilibrium is preserved. Hold the population fixed and replace the mechanism: the equilibrium changes. Behaviour traceable to the population alone is a small residual on top of architecture-driven dynamics.*

Claim 1 is identification of the dominant explanatory variable. Claim 2 is the operational counterfactual that licenses Claim 1: the population is held fixed and the architecture is varied. *Ghosting Isn't Cowardice* states the counterfactual explicitly — replacing the user pool with people from 1995 under the same dating-app architecture produces the same equilibrium — and the move recurs across the corpus.

2. Why this scale. Classical natural science has measured the scales below the human (atoms, cells, neurons) and above the human (climate, markets-as-aggregate, populations) to high resolution. The scale *between* two or N agents interacting through an architecture is, by comparison, methodologically nearly empty. There is no canonical formal language for it. The most-cited models live in microeconomics (game theory) and computer science (mechanism design) but are rarely applied outside their disciplines, and the phenomena most affected by inter-agent dynamics — platform behaviour, dating, peer review, attention extraction, identity signalling — are typically left to qualitative discourse.

The marginal return on understanding one more electron is, for the purposes of subjective experience, near zero. The marginal return on formalising one previously unmodelled inter-agent mechanism is large, because the prior is not “well-established theory needs an additional decimal place” but “no formal theory exists.” This is the slot the corpus operates in. It is also the structural reason that pieces of mathematics typically considered elementary — a multiplicative cascade, a free-energy bound, an eigenvector asymmetry — yield real explanatory traction in this regime: the comparison is not against tighter models but against no model at all.

A note on what is and is not new. The argument that inter-agent structure carries explanatory weight is not new in any single discipline; it is the operating assumption of mechanism design, of much of sociology, of evolutionary game theory. What is being claimed is narrower. First, that across these disciplines the move is consistently localised to its native vocabulary, so the same structural lemma — a threshold game, say — gets re-derived in three places without recognition. Second, that many of the phenomena that fill ordinary life (ghosting, burnout, identity broadcast on platforms, the felt arbitrariness of recommendation) sit between disciplines and are therefore picked up by none. The corpus reads across these boundaries.

3. Modelling under observer effects. The standard objection to formalising inter-agent mechanisms is reflexive: publishing a model of σ_R changes how people ghost; publishing the platform food chain changes how creators read their own role; the act of measurement is itself the perturbation. The objection is real. It is not grounds to refuse the work, because *impossibility* is not a single condition — it has facets, and each admits a different attack.

Remark 1 (Three modes of impossibility).

<i>Mode</i>	<i>Difficulty</i>	<i>Attack</i>
<i>Information-theoretic</i>	<i>Capturing the full state would require infinite or impossibly clean data</i>	<i>Reduce to a sufficient statistic; the relevant scalar can be finite even when the underlying process is high-dimensional</i>
<i>Reflexive</i>	<i>The model perturbs the system being modelled</i>	<i>Publish the model jointly with the architectural change that defines the new equilibrium, so the perturbation is directed rather than incidental</i>
<i>Conceptual</i>	<i>No agreed-upon target object exists</i>	<i>Show that the missing scalar ground truth was never the right object; supply the structured object that does the work</i>

Information-theoretic. *Cartel or Creative Destruction?* faces precisely this difficulty: standard concentration indices are degenerate across two welfare-opposite regimes (collusive cartel and Schumpeterian innovation-driven monopoly). The attack is to identify a sufficient statistic — the eigenvector

tor asymmetry of the residual price covariance — that separates the regimes with Cohen’s $d > 70$. The scalar is finite even though the underlying market state is not.

Reflexive. *Ghosting Isn’t Cowardice* models σ_R and immediately proposes the architectural lever (e.g. Hinge’s “Your Turn” feature) that raises c_{silence} and flips the equilibrium. The model is not a passive observation but a directed perturbation: it ships with the change that justifies it. *The TikTok Peer Review* does the analogous move at second order, modelling the mechanism that produces models. In both cases reflexivity is acknowledged and converted into design.

Conceptual. Personality Is Not Who You Are. It Is Your Cost Tensor is the cleanest case. The Big Five framework presupposes a five-dimensional scalar object whose component values describe a person stably across context. The cost-tensor model shows this object cannot exist as specified, because the relevant quantity is context-dependent in a way no projection Π_5 can preserve. The apparent unmeasurability of personality is a misspecification of the target object, not a fundamental limit of measurement. Once the object is replaced by a cost tensor under a context operator L_W , it is identifiable; the intervention mechanism (operator sequences with eigenvalues less than one) is also identified.

4. What this program is and is not. The program is not a claim that subjective experience reduces to mechanism. The cost-tensor paper itself acknowledges Λ_{trauma} as a strictly individual operator with at least one eigenvalue $\gg 1$; some variance is genuinely person-shaped. The claim is narrower: that the variance commonly attributed to character is largely architectural, and that the architectural component is the under-modelled component. “Architecture > population” is a statement about marginal explanatory power, not about ontology.

The program is also not a claim that all formalisation is equally tractable. Cartel detection admits a clean sufficient statistic; subjective wellbeing under social comparison probably does not. The three modes of impossibility are not a guarantee that every reflexive system can be modelled, only that the standard rejection (“you will perturb it, therefore do not”) over-generalises a real local difficulty into a fictitious global one.

What unites the corpus, then, is not a domain (which ranges from drug discovery to dating apps) and not a single mathematical object (operators, multiplicative cascades, free-energy bounds, threshold games all appear), but a methodological commitment: locate the dominant variable in the architecture, not the population; treat reflexive perturbation as something to be designed rather than avoided; and, where the standard target object is conceptually misspecified, supply the right one rather than concede unmeasurability.

5. Status. This is a draft (v0.1) written in conversation with the corpus, not in front of it. It will be revised as the corpus grows and as the connections between pieces become load-bearing. The next obvious extensions are (i) a structural lemma explicitly unifying the threshold-bifurcation skeleton across *Burnout*, *Ghosting*, *ADHD Attractor*, and the *Cost Tensor*; (ii) a precise statement of the “architecture-not-population” identification result, with its scope and failure modes; (iii) a fourth mode of impossibility, computational, for systems where a tractable model exists in principle but not in practice.

Cited works in this corpus. *Burnout Is Not Too Much Work, v2* (2026-05-06). *Ghosting Isn’t Cowardice. It’s Math.* (2026-04-24). *ADHD Is Not a Deficit. It’s a Different Attractor.* (2026-04-24). *Personality Is Not Who You Are. It Is Your Cost Tensor.* (2026-05-06). *Cartel or Creative Destruction? Eigenvector Asymmetry as a Spectral Diagnostic for Schumpeterian Market Structure* (2026-04-22). *The TikTok Peer Review* (2026-04-26). *The Platform Has a Food Chain. You Are In It.* (2026-05-06). All available at hakvinv.github.io/h.vosteen-research/.